Retaining Knowledge and Enhancing Long-Text Representations in CLIP through Dual-Teacher Distillation

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Abstract

Contrastive language–image pretraining models such as CLIP have demonstrated remarkable performance in various text-image alignment tasks. However, the inherent 77-token input limitation and reliance on predominantly short-text training data restrict its ability to handle long-text tasks effectively. To overcome these constraints, we propose LongD-CLIP, a dual-teacher distillation framework designed to enhance long-text representation while mitigating knowledge forgetting. In our approach, a teacher model, fine-tuned on long-text data, distills rich representation knowledge into a student model, while the original CLIP serves as a secondary teacher to help the student retain its foundational knowledge. Extensive experiments reveal that LongD-CLIP significantly outperforms existing models across long-text retrieval, short-text retrieval, and zero-shot image classification tasks. For instance, in the image-to-text retrieval task on the ShareGPT4V test set, LongD-CLIP exceeds Long-CLIP's performance by 2.5%, achieving an accuracy of 98.3%. Similarly, on the Urban-1k dataset, it records a 9.2% improvement, reaching 91.9%, thereby underscoring its robust generalization capabilities. Additionally, the text encoder of LongD-CLIP exhibits reduced latent space drift and improved compatibility with existing generative models, effectively overcoming the 77-token input constraint.

1. Introduction

Recent advancements in vision-language models have significantly propelled multimodal learning, enabling more precise alignment between images and textual descriptions. Among these models, CLIP (Contrastive Language-Image Pretraining) [20] has emerged as a powerful approach due to its unique ability to perform zero-shot learning [7, 23, 24],

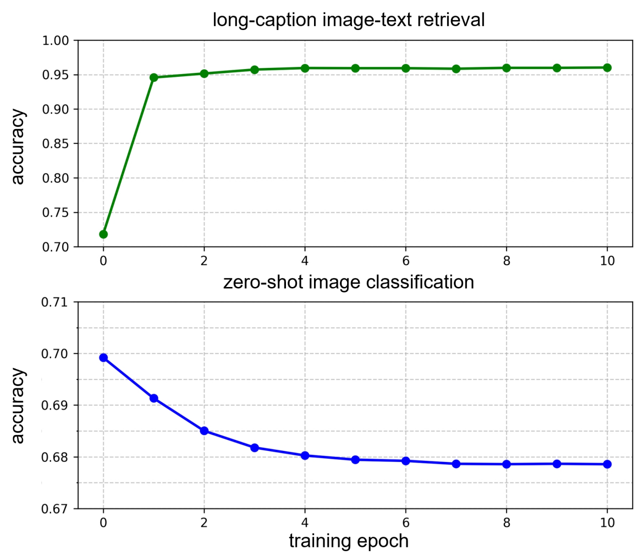

Figure 1. Performance curves for long-text retrieval and zero-shot classification during CLIP fine-tuning on the long-text image dataset ShareGPT4V. Values represent the average across our test datasets.

effectively transferring knowledge across a wide range of tasks. This versatility spans applications from image classification [1] to more complex tasks like image-text retrieval [17], where the model demonstrates strong adaptability to varied content. CLIP's approach centers on a contrastive learning objective, which facilitates the learning of joint representations of images and concise text descriptions, such as captions or short phrases. Through this mechanism, CLIP has achieved robust performance in associating images with short text prompts. However, despite its strength with brief descriptions, CLIP's effectiveness diminishes when dealing with longer detailed text inputs.

Long-text inputs challenge models like CLIP, which are mainly trained on short descriptions. The DCI dataset [25] allows evaluation of vision-language models on detailed,

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mask-aligned visual descriptions, serving as a fine-grained benchmark to reveal the strengths and weaknesses of CLIP-style models on densely annotated data. Another critical challenge is CLIP's text encoder, which uses a learnable absolute positional embedding capped at 77 tokens, limiting input length. This fixed-length tokenization can lose important semantic details in longer texts, reducing the model's ability to grasp extended narratives. Recent work, Long-CLIP [32], shows that only the first 20 positional embeddings in CLIP's text encoder are effectively trained. To overcome this, Long-CLIP linearly interpolates the remaining embeddings, expanding the input capacity to 248 tokens for more comprehensive long-text processing.

However, while fine-tuning enhances CLIP's capability in representing long text, an unintended phenomenon—knowledge forgetting—emerges. Specifically, as the model's long-text retrieval performance improves, its zero-shot classification accuracy deteriorates. This trade-off indicates that augmenting one ability may inadvertently lead to the erosion of critical, previously acquired knowledge. As illustrated in Figure 1, the performance curves of long-text retrieval and zero-shot classification exhibit a clear inverse relationship during fine-tuning. In our experiments, we extended CLIP's input limit and trained it for 10 epochs on the ShareGPT4V dataset [3], evaluating both long-text retrieval and zero-shot image classification after each epoch. The results reveal that while long-text retrieval benefits, the image classification accuracy declines by approximately 2%. These findings are based on averages across multiple test datasets: long-text performance was measured on ShareGPT4V [3] and Urban-1k [32], and zero-shot image classification was evaluated on ImageNet-1K [5], ImageNet-V2 [21], ImageNet-O [9], CIFAR-10 [11], and CIFAR-100 [11].

To tackle these challenges, we propose a dual-teacher network distillation approach that enhances CLIP's long-text representation while mitigating knowledge forgetting. First, we extensively fine-tune a CLIP model on a long-text image dataset, thereby creating a teacher network with a strong ability to capture detailed long-text and image information. Although this teacher may suffer from some degree of knowledge forgetting, it excels in representing long-text semantics. Next, we employ knowledge distillation to train a student network using both the long-text-enhanced teacher and the original CLIP model as instructors. This dual-teacher framework allows the student to inherit the long-text proficiency from the fine-tuned teacher while preserving the broad, foundational knowledge of the original model, effectively achieving a balanced performance.

The contributions of this paper are threefold:

(1) **Enhanced long-text representation**: We significantly improve CLIP's ability to handle long-text inputs, achieving superior performance in retrieval tasks requiring detailed comprehension.

(2) **Reduced knowledge forgetting**: Our approach minimizes knowledge forgetting, preserving zero-shot classification accuracy while enhancing long-text processing capabilities.

(3) **Improved latent space alignment**: We mitigate latent space drift in text embeddings, ensuring better compatibility with image generation models.

2. Related Work

2.1. VLMs Knowledge Distillation

Most VLM knowledge distillation methods focus on adapting image-level representations to region- or pixel-level tasks, such as object detection [4, 6, 8, 28] and semantic segmentation [15, 29, 35]. For example, ViLD [8] distills VLM knowledge into a two-stage detector by aligning its embedding space with that of the CLIP image encoder, while CLIPSeg [16] uses a lightweight transformer decoder to extend CLIP for segmentation tasks. LSeg [12] enhances segmentation by correlating CLIP text embeddings with pixel-wise embeddings from segmentation models. ZegCLIP [35] leverages CLIP to produce semantic masks, incorporating a relationship descriptor aimed at mitigating overfitting on base classes. MaskCLIP+ [34] and SSIW [31] use pixel-level pseudo labels predicted by vision-language models (VLMs) as a basis for knowledge distillation. FreeSeg [19] initially generates mask proposals, then performs zero-shot classification on each mask individually.

These methods rely on both image and text embeddings to effectively transfer CLIP's pre-trained knowledge to downstream tasks. We introduce a dual-teacher distillation strategy to overcome CLIP's long-text representation limitations, maintaining zero-shot generalization while enhancing long-text processing for diverse downstream tasks.

2.2. Contrastive Vision-Language Models with Long Captions.

Contrastive vision-language models like CLIP [20], ALIGN [10], and BLIP [13] have significantly advanced visual-text alignment by leveraging large-scale contrastive learning, achieving strong performance in tasks like image-text retrieval and zero-shot classification [26, 27]. However, these models are typically limited to short, global text descriptions due to restricted context windows, such as CLIP's 77-token limit.

Building on this insight, DCI [25] proposes that "A Picture is Worth More Than 77 Text Tokens" and introduces the Densely Captioned Images (DCI) dataset, comprising 7805 natural images with human-annotated, mask-aligned descriptions that average over 1000 words each. Additionally, it evaluates CLIP-style models on these dense captions. DreamLIP [33] uses a pre-trained Multi-modality

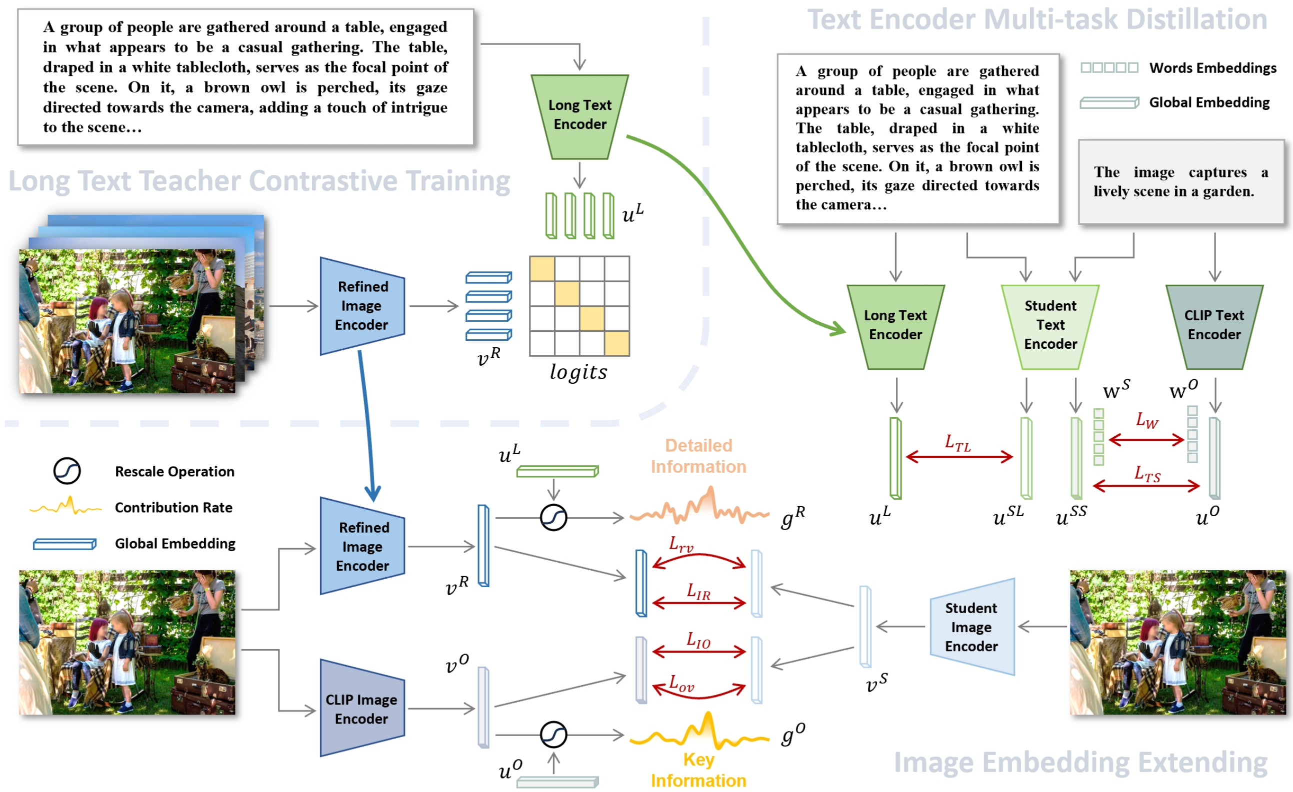

Figure 2. Overall framework of our two-stage dual-teacher distillation method. First, a highly performant CLIP model is fine-tuned on long-text and image data to serve as the teacher model. Then, the text and image encoders are distilled separately. The student text encoder is trained to handle both long-text and short-text tasks, while the student image encoder learns the teacher model's ability to capture fine details, with measures in place to prevent knowledge forgetting.

Large Language Model (MLLM) to generate detailed captions for 30M images and investigates these captions within a contrastive learning framework. It dynamically selects sub-captions from text labels to create multiple positive pairs, aligning each sub-caption's embedding with its related local image patches in a self-supervised manner. With training on 30M image-text pairs, DreamLIP matches or even outperforms CLIP, which was trained on 400M pairs. However, DreamLIP's latent space is distinct from CLIP's, which restricts its compatibility with pre-trained modular CLIP components in downstream tasks. Long-CLIP [32] modifies CLIP by extending the absolute position encodings to 248 through interpolation and fine-tuning on long captions. However, Long-CLIP's primary component matching approach limits the model's ability to learn effectively from long text data and leads to slight knowledge forgetting.

Our approach tackles the issue of long-text representation in CLIP by employing a dual-teacher distillation strategy, which maintains CLIP's zero-shot generalization capabilities while enhancing its ability to process long-text inputs effectively.

3. Method

First, we extensively train the CLIP model on a long text-image dataset, designating it as the teacher network in our distillation process. While this teacher model may experience some degree of knowledge forgetting, it demonstrates strong capabilities in representing long text and capturing intricate image details. We then perform knowledge distillation, enabling the student network to learn from both the long-text-enhanced teacher model and the original CLIP model, thereby enhancing the student model's overall performance.

3.1. Long Text Teacher Network Training

In the first step of our method, we train a long-text version of the CLIP model, which will serve as the teacher network in subsequent steps. This process begins by initializing the model with weights from the original CLIP. The training phase relies solely on contrastive learning to fully adapt the model. Notably, the original CLIP supports only 77-token text inputs. To extend the input length, we apply the same initialization approach as Long-CLIP, retaining the first 20 positional embeddings and using linear interpolation of the positional embeddings to extend the text encoder's maximum input length to 248 tokens.

```
# image_encoder - Vision Transformer
# text_encoder - Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# extract image and text features
I_f = image_encoder(I)  # [n, d_i]
T_f = text_encoder(T)  # [n, d_t]
# normalize image and text embeddings
I_e = I_f / I_f.norm(dim=1, keepdim=True)
T_e = T_f / T_f.norm(dim=1, keepdim=True)
# get batchsize
batch_size = I_e.shape[0]
# compute cosine similarities
logits_per_i = torch.matmul(I_e, T_e.T)
logits_per_i = logits_per_i * (logit_scale.exp())
logits_per_t = logits_per_i.T
# compute loss
labels = torch.arange(batch_size)
loss_i2t = F.cross_entropy(logits_per_i, labels)
loss_t2i = F.cross_entropy(logits_per_t, labels)
loss = (loss_i2t + loss_t2i) / 2
```

Figure 3. Torch-like pseudocode for contrastive learning implementation of long-text teacher model.

In Figure 3, we present the pseudocode related to contrastive learning. Through contrastive learning with long-text inputs, the model learns to capture fine-grained information within both long texts and images. The contrastive training process is identical to that of CLIP [9].

3.2. Dual-teacher Distillation Learning

At this stage, we have two models: one with enhanced long-text representation capabilities and the original model, which retains extensive pre-trained knowledge. We use both models as teacher networks in our knowledge distillation process. By leveraging the unique characteristics of the text encoder (Transformer) and image encoder (ViT), we apply tailored distillation strategies for each.

3.2.1. Text Encoder Multi-task Distillation

To produce robust embeddings for both long-text and short-text tasks, we train the student network's text encoder via knowledge distillation using two teacher networks: the long-text CLIP encoder, which strengthens the student's long-text representation, and the original CLIP text encoder, which preserves the base model's knowledge.

As shown in Figure 2, the long-text embedding $\mathbf{u}^L$ is from the long text encoder, while the short-text embedding $\mathbf{u}^O$ is from the CLIP text encoder. The student text encoder receives short and long texts, outputting embeddings $\mathbf{u}^{SS}$ and $\mathbf{u}^{SL}$, which are trained to align with $\mathbf{u}^O$ and $\mathbf{u}^L$, respectively.

To construct the distillation process for the text encoder, we use cosine similarity loss, defined as $1 - \langle \cdot, \cdot \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the cosine similarity between two vectors. Let $\mathcal{L}_{TS}$ and $\mathcal{L}_{TL}$ represent the distillation losses for the short-text and long-text tasks, respectively.

$$\mathcal{L}_{TS} = 1 - \langle \mathbf{u}^O, \mathbf{u}^{SS} \rangle, \tag{1}$$

$$\mathcal{L}_{TL} = 1 - \langle \mathbf{u}^L, \mathbf{u}^{SL} \rangle. \tag{2}$$

To minimize latent space drift in the text encoder, we impose a feature constraint on the first 20 word-token embeddings from the original and student text encoders, denoted as $\mathbf{w}^O$ and $\mathbf{w}^S$. This follows Long-CLIP's observation that CLIP's effective representation length is about 20 tokens. Let $\mathcal{L}_W$ denote the distillation loss for word tokens:

$$\mathcal{L}_W = \|\mathbf{w}^O - \mathbf{w}^S\|^2. \tag{3}$$

The distillation loss applied to the student text encoder is represented as follows:

$$\mathcal{L}_T = \mathcal{L}_{TS} + \mathcal{L}_{TL} + \mathcal{L}_W \tag{4}$$

The student model's text encoder is initialized with the original CLIP weights, while the position embeddings are extended through interpolation. In our experiments, multi-task distillation allows the student encoder to converge well on both short-text and long-text tasks.

3.2.2. Image Embedding Extending

While the refined image encoder in the long-text teacher network captures fine details well, it experiences knowledge forgetting. Directly distilling from it risks passing this issue to the student network, so we adopt a dual-teacher distillation strategy. Unlike text encoder distillation, both image encoder teachers receive the same input. Our goal is for the student network to retain the knowledge from the original CLIP image encoder while gaining the detailed representation abilities of the long-text version's image encoder.

The output of the student image encoder is $\mathbf{v}^S$. We aim to learn the key information from the original CLIP image embedding $\mathbf{v}^O$ and the detailed information from the long-text version CLIP image embedding $\mathbf{v}^R$. For a given image embedding, the importance of each dimension can be measured by its contribution to cross-modal similarity:

$$\cos\langle \mathbf{u}, \mathbf{v} \rangle = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \tag{5}$$

We define an importance function, denoted by g, which takes the image embedding $\mathbf{v}$ and the text embedding $\mathbf{u}$ as inputs. Here, i refers to the ith dimension of the embedding:

$$g(\mathbf{u}_i, \mathbf{v}_i) = \exp(t \cdot \frac{\mathbf{u}_i}{\|\mathbf{u}\|} \cdot \frac{\mathbf{v}_i}{\|\mathbf{v}\|}). \tag{6}$$

Figure 4. Long-Text to Image Retrieval Results. Our model retrieves images with more accurately matched details. The two examples shown are from the Urban-1k dataset and the ShareGPT4V validation set, respectively.

Here, t represents the temperature coefficient. With the dimension-wise re-weighting coefficient g_i (where $i \in [1, N]$) applied to each element in the N-dimensional embedding space, the distillation loss functions for the original model's image encoder $\mathcal{L}_{\text{ov}}$ and the refined image encoder $\mathcal{L}_{\text{rv}}$ can be formally defined as follows:

$$\mathcal{L}_{ov} = \frac{1}{N}\sum_i((\mathbf{v}_i^O - \mathbf{v}_i^S)^2 \cdot g_i(\mathbf{u}_i^O, \mathbf{v}_i^O)), \tag{7}$$

$$\mathcal{L}_{rv} = \frac{1}{N}\sum_i((\mathbf{v}_i^R - \mathbf{v}_i^S)^2 \cdot g_i(\mathbf{u}_i^L, \mathbf{v}_i^R)). \tag{8}$$

To ensure stable training, cosine similarity loss is added. Let $\mathcal{L}_{IO}$ and $\mathcal{L}_{IR}$ represent the distillation losses for the original CLIP image encoder and the refined image encoder, respectively.

$$\mathcal{L}_{IO} = 1 - \langle \mathbf{v}^O, \mathbf{v}^S \rangle, \tag{9}$$

$$\mathcal{L}_{IR} = 1 - \langle \mathbf{v}^R, \mathbf{v}^S \rangle. \tag{10}$$

The total loss function for training the student image encoder can be expressed as follows, where λ_1, λ_2, λ_3 and λ_4 represent the respective weighting coefficients:

$$\mathcal{L}_I = \lambda_1\mathcal{L}_{ov} + \lambda_2\mathcal{L}_{rv} + \lambda_3\mathcal{L}_{IO} + \lambda_4\mathcal{L}_{IR}. \tag{11}$$

Through our distillation strategy, the student image encoder gains the ability to capture fine image details while retaining original knowledge. Additionally, our designed re-weighting strategy supervises both long-text and short-text matching effectively.

4. Experiments

4.1. Experiment Setting

In our experiments, we assess model performance on three tasks: long-text and short-text retrieval to evaluate retrieval capabilities, and zero-shot image classification to detect knowledge forgetting in unfamiliar categories. The datasets used are as follows:

Long-Caption Image-Text Retrieval. For long-caption image-text retrieval, we adopt the approach of Long-CLIP [32], using a random sample of 1k image and long-text pairs from ShareGPT4V [3] and Urban1K [32]. The Urban1K dataset includes 1k images depicting urban scenes, for which long captions were generated using GPT-4V [2].

short-caption image-text retrieval. For traditional short-caption image-text retrieval, we follow the approach of Long-CLIP [32], utilizing COCO2017 [14] and Flickr30k [30]. The 5k validation set is used for COCO2017, while for Flickr30k, we employ the full 30k dataset instead of the typical 1k test set.

zero-shot image classification. The primary dataset used is ImageNet-1K [5]. Additionally, ImageNet-V2 [21], ImageNet-O [9], CIFAR-10 [11], and CIFAR-100 [11] are included to further analyze classification capability.

4.2. Implement Detail

Our model training includes two architectures: ViT-B/16 and ViT-L/14. Training is conducted solely on the ShareGPT4V dataset, with evaluation performed on other datasets. For fine-tuning the long-text teacher model, we use a batch size of 1024 and train for 10 epochs. During

Table 1. The R@1 of long-caption text-image retrieval on 1k ShareGPT4V validation set and Urban-1k dataset. Best result is in **bold**.

		ShareGPT4V		Urban-1k	
		Image-to-Text	Text-to-Image	Image-to-Text	Text-to-Image
B/16	CLIP	78.2	79.6	68.1	53.6
	Long-CLIP	94.6	93.3	78.9	79.5
	Ours	**99.0**	**98.4**	**87.2**	**87.3**
L/14	CLIP	81.8	84.0	68.7	52.8
	Long-CLIP	95.8	95.6	82.7	86.1
	Ours	**98.3**	**98.6**	**91.9**	**90.8**

Table 2. Short-caption text-image retrieval on the 5k COCO2017 validation set and the whole 30k Flickr30K dataset. Best result is in **bold**.

		COCO						Flickr30k					
		Image-to-Text			Text-to-Image			Image-to-Text			Text-to-Image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
B/16	CLIP	51.8	76.8	84.3	32.7	57.7	68.2	44.1	68.2	77.0	24.7	45.1	54.6
	Long-CLIP	57.6	81.1	87.8	40.4	65.8	75.2	46.8	71.4	79.8	34.1	56.3	65.7
	ours	**59.4**	**82.7**	**89.5**	**41.1**	**67.2**	**77.1**	**52.9**	**76.0**	**83.7**	**34.5**	**57.6**	**67.0**
L/14	CLIP	56.1	79.5	86.8	35.4	60.1	70.2	48.5	72.6	80.8	28.0	49.3	58.7
	Long-CLIP	62.8	85.1	91.2	46.3	70.8	79.8	53.4	77.5	85.3	41.2	64.1	72.6
	Ours	**64.4**	**85.2**	**91.5**	**47.0**	**72.0**	**80.8**	**60.2**	**82.0**	**88.4**	**42.2**	**65.5**	**74.1**

Table 3. Results of zero-shot image classification in the above five validation sets. Best result is in **bold**.

		ImageNet	ImageNet-O	ImageNet-V2	Cifar10	Cifar100	Average
B/16	CLIP	**68.4**	42.2	**61.9**	90.8	67.3	66.12
	Long-CLIP	66.8	**42.7**	61.2	90.7	69.3	66.14
	ours	68.1	41.4	61.7	**92.0**	**69.9**	**66.62**
L/14	CLIP	**75.5**	31.9	**69.9**	95.5	76.8	69.92
	Long-CLIP	73.5	**33.7**	67.9	95.3	78.5	69.78
	Ours	75.2	33.3	68.8	**96.1**	**79.4**	**70.56**

distillation, we train for 5 epochs with a smaller batch size of 64, as this configuration enhances distillation effectiveness and improves generalization across different data distributions. The same gradient update settings are used for both training phases. A learning rate of 1×10^{-6} is applied, with a warm-up phase during the first 200 iterations to stabilize the learning process. A weight decay of 1×10^{-2} is employed to regularize the model. The AdamW optimizer is configured with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$, ensuring stable and efficient convergence. The temperature coefficient t is set to 25, and the regularization weights are $\lambda_1 = 0.5$, $\lambda_2 = 0.5$, $\lambda_3 = 1$, and $\lambda_4 = 2$.

4.3. Results and Analysis

Our experimental results show that the dual-teacher distillation strategy enhances CLIP's long-text representation while also improving short-caption retrieval. The model's performance in zero-shot classification demonstrates its ability to retain CLIP's original knowledge, allowing adaptability to diverse downstream tasks.

Long-Caption Text-Image Retrieval

As quantitatively demonstrated in Table 1, our model exhibits significant improvements in long-caption text-image retrieval tasks on both ShareGPT4V and Urban-1k datasets. Notably, under the ViT-B/16 configuration, our model reaches state-of-the-art R@1 scores of 99.0 and 98.4 for Image-to-Text and Text-to-Image tasks, respectively, surpassing both Long-CLIP (94.6 and 93.3) and the original CLIP (78.2 and 79.6). For the ViT-L/14 architecture, our model also achieves R@1 scores of 98.3 and 98.6 on the ShareGPT4V dataset, further improving upon the performance of the long-text version of CLIP. On the Urban-1k dataset, the Image-to-Text task with ViT-L/14 shows a 9.2% improvement, demonstrating our model's strong generalization for long-text descriptions in complex urban scenarios. These results indicate that our approach effectively enhances CLIP's long-text representation capability while maintaining sensitivity to image details, which is essential for open-world retrieval applications, particularly in domain-specific scenarios with rich contextual descriptions.

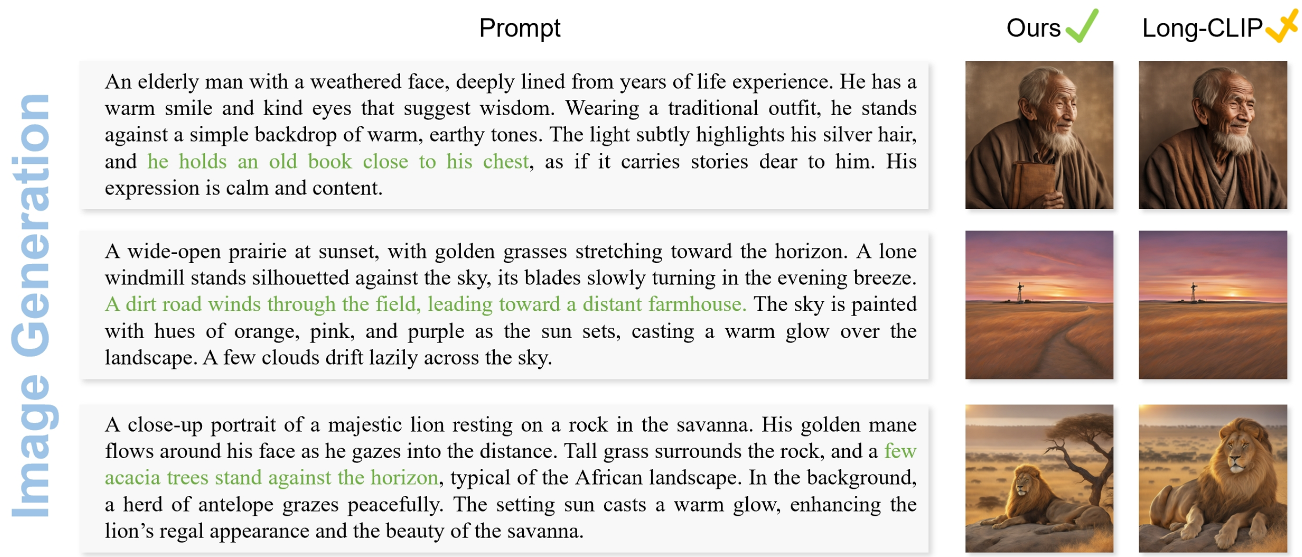

Figure 5. Plug-and-Play in Image Generation Model (SDXL). Our text encoder demonstrates better compatibility with SDXL, ensuring the preservation of detailed information in generated images.

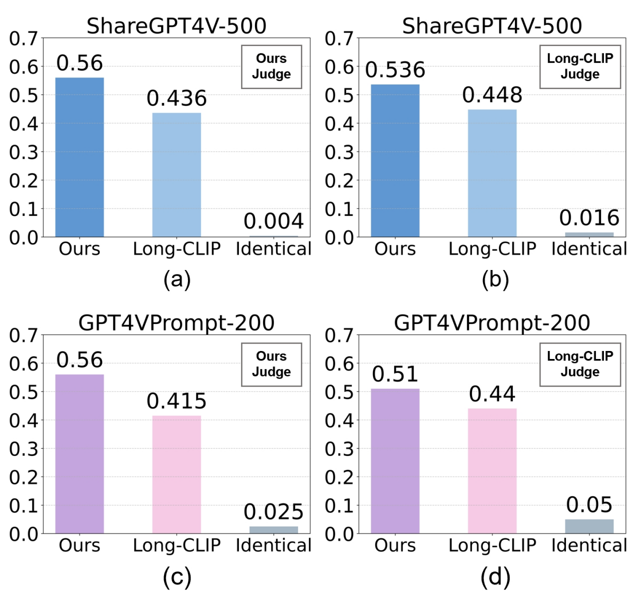

Figure 6. SDXL Text Encoder Replacement Experiment. Comparison of CLIP Scores between our SDXL and Long-CLIP's SDXL on two sets of prompts. The bar values indicate the proportion of cases where each model achieved a higher CLIP Score, with "Identical" representing cases with equal scores.

Short-Caption Text-Image Retrieval

In the short-caption text-image retrieval task, our model also performs well on the COCO2017 and Flickr30k datasets (see Table 2). For the ViT-B/16 architecture on the COCO dataset, our model achieves R@1 scores of 59.4 and 41.1 for Image-to-Text and Text-to-Image tasks, respectively, outperforming both Long-CLIP and the original CLIP. The results on the Flickr30k dataset are similar, with our approach reaching R@1 scores of 52.9 and 34.5 for Image-to-Text and Text-to-Image tasks, respectively, again surpassing other baseline models. For the ViT-L/14 architecture, our model achieves R@1 scores of 64.4 (Image-to-Text) and 47.0 (Text-to-Image) on the COCO dataset, showing a substantial improvement over Long-CLIP. On the Flickr30k dataset, our method also achieves the best results for both R@1 and R@10 metrics. This demonstrates that our dual-teacher distillation strategy significantly improves retrieval performance, even in short-caption tasks.

Zero-Shot Image Classification

In the zero-shot image classification task, our model exhibits strong generalization and knowledge retention capabilities (see Table 3). On the ImageNet, ImageNet-V2, CIFAR-10, and CIFAR-100 datasets, our model performs well across most benchmarks, particularly achieving an average score of 70.56 with the ViT-L/14 architecture, significantly surpassing the original CLIP and Long-CLIP. This result verifies that our approach effectively mitigates knowledge forgetting, enabling robust adaptability across diverse data distributions.

4.4. Plug-and-Play in Image Generation

In text-to-image generation, models like Stable Diffusion [22] use a pre-trained text encoder, but the original CLIP text encoder is limited to 77 tokens. Long-CLIP overcomes this limitation with a plug-and-play capability. Our model can be integrated into text-to-image generation models like

Table 4. Ablation Study. Comparison of our method with other training and fine-tuning approaches. I2T and T2I refer to Image-to-Text Retrieval and Text-to-Image Retrieval, respectively, and CL represents contrastive learning.

Cross-modal Retrieval Task		ShareGPT4V		Urban-1k		COCO		Flickr30k	
		I2T	T2I	I2T	T2I	I2T	T2I	I2T	T2I
L/14	CLIP	81.8	84.0	68.7	52.8	56.1	35.4	48.5	28.0
	O-teacher + CL	97.3	97.1	81.5	83.2	57.9	37.2	49.7	29.9
	L-teacher	99.2	99.1	92.7	92.1	64.7	47.3	55.0	42.5
	Ours	98.3	98.6	91.9	90.8	64.4	47.0	60.2	42.2

Image Classification Task		ImageNet	ImageNet-O	ImageNet-V2	Cifar10	Cifar100	Average
L/14	CLIP	75.5	31.9	69.9	95.5	76.8	69.92
	O-teacher + CL	75.5	31.3	69.7	95.7	77.5	69.94
	L-teacher	70.8	32.4	64.9	94.9	76.3	67.86
	Ours	75.2	33.3	68.8	96.1	79.4	70.56

Long-CLIP, offering better compatibility due to reduced latent space drift in text embeddings.

In Figure 5, we show three sets of images generated by SDXL [18] using long-text prompts. Compared to Long-CLIP, our text encoder adapted for SDXL can capture more detailed information, such as the book in an old man's hand, the road in a field, and the tree behind a lion.

Additionally, we performed quantitative comparisons by randomly sampling 500 texts from ShareGPT4V (blue bars in Figure 6) and 200 long-text prompts from GPT-4 (pink bars) for the image generation experiment. With the same input and random seed, we used our model and Long-CLIP as the text encoder for SDXL to generate images and then calculated the CLIP Score for each generated image. The CLIP Score is defined as the cosine similarity between image and text embeddings.

By comparing CLIP Scores, we determine which text encoder is better suited to SDXL. For fairness, the CLIP Score was calculated using both our model and Long-CLIP, and the comparison results were converted to ratios in Figure 6. The results indicate that our method outperforms Long-CLIP regardless of whether our model or Long-CLIP is used as the judge. Further analysis reveals that using our model as the judge leads to fewer "Identical" cases, indicating greater sensitivity to image details.

4.5. Ablation Study

In Table 4, we compare two training strategies. The first approach (**O-teacher + CL**) uses contrastive learning with an embedding constraint based on the original CLIP output. While this method mitigates knowledge forgetting in zero-shot tasks, it significantly restricts the model's ability to learn effective long-text representations. The second approach, **L-teacher**, is our long-text teacher model, which is thoroughly trained on the ShareGPT4V dataset and exhibits strong long-text retrieval capabilities. However, it suffers from knowledge forgetting, with an average drop of approximately 2% in zero-shot image classification accuracy, indicating a degradation in generalization.

In contrast, our two-stage distillation strategy demonstrates clear improvements across both tasks. Notably, on the Flickr30k dataset, our method surpasses the long-text teacher model by 5.2% and the original CLIP by approximately 11.7% in Image-to-Text retrieval. This result suggests that our approach successfully unlocks the model's latent potential and enhances its representation capabilities.

Table 5. Ablation study on the distillation loss of the student image encoder. I2T and T2I refer to Image-to-Text Retrieval and Text-to-Image Retrieval. Only the average values are shown for Zero-shot Image Classification.

ViT-L/14	ShareGPT4V		Urban-1k		Zero-Shot Image Classification
	I2T	T2I	I2T	T2I	
MSE	96.0	95.9	83.4	84.1	69.90
Cosine	97.2	96.5	88.3	86.7	70.35
Ours	98.3	98.6	91.9	90.8	70.56

In our experiments, the text encoder converges well in both long-text and short-text distillation. In Table 5 compared to using only MSE loss or cosine similarity loss, our distillation strategy for the student image encoder achieves better results.

5. Conclusion

We present a dual-teacher distillation framework that systematically enhances CLIP's capabilities in cross-scale text retrieval (long/short), zero-shot recognition, and cross-modal generation compatibility. These consistent gains demonstrate the model's improved long-text capabilities and its retention of valuable knowledge, laying a foundation for future multimodal learning advancements.

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